Lesson plan · for the teacher

Models that remember too much

AI security · Level 3 · Lesson 2

Time	Materials
40 min	A made-up class register on paper, with names, phone numbers, addresses and notes (one copy per group); Dark pencils or markers; Board and chalk

Learning goals

- Explain memorisation and membership inference.
- Remove identifiers from text before it is used.
- Weigh privacy against usefulness, and ask the right questions.

Activity: Black it out

1. Warm-up (5 min): Show the made-up register. Ask: if a stranger saw this, what could they learn about each pupil? List answers on the board.
2. Teach (8 min): Explain that an AI model trained on text can store rare lines and repeat them. Agree what an identifier is: names, phone and ID numbers, addresses, and rare details that point to one person.
3. Black it out (12 min): Each group blacks out every identifier on its copy with a dark pencil.
4. Swap and check (8 min): Groups swap copies and circle anything the other group missed. Share the best catch with the class.
5. Wrap-up (7 min): Agree one rule for what the school never puts into an AI tool and write it on the board. Then give the exit ticket.

Use only made-up names and numbers, never a real register. Hide two or three rare details (“the only pupil who walks from the river village”) so groups learn that identifiers are not only names and numbers.

Discussion

- Which identifiers were hardest to spot, and why?
- Why can't we just delete the file after a model has learned from it?
- When is it worth making a model less useful to protect privacy?

Exit ticket

- Write three kinds of identifier, and one question to ask before any pupils' data goes to an AI service.

Try it in class tomorrow

Play “Black it out”: give groups a made-up class register with names, phone numbers and notes. Groups black out every identifier by hand, then swap with another group and look for anything the others missed.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A company offers a free marking assistant if the school uploads every pupil's past essays to train it. What should the school ask first?**

◯ How fast does it mark? ◯ What data does it need, will it train on it, and can it be deleted?

◯ Can it mark in our language?

2 **Finish redact(text) so it replaces every digit with "#" and keeps everything else. The example line should print as Call #### ### #### after #pm.**

```
def redact(text):
    out = ""
    for c in text:
        # add "#" for a digit, else c
        out += c
    return out

print(redact("Call 0803 555 1234 after 4pm"))
```

Write your program:

3 **This redact() should turn "Room 12" into "Room ##", but it returns just "#". Which line has the mistake?**

Circle the line with the mistake. Write it correctly:

```
1  def redact(text):
2      out = ""
3      for c in text:
4          if c.isdigit():
5              out = "#"
6          else:
7              out += c
8      return out
```

4 **Which of these are identifiers you should remove before pupils' text is used to train a model? Choose all that apply.**

- [ ] A pupil's full name
- [ ] A parent's phone number
- [ ] “The only pupil in 5B with twins at home”
- [ ] “Photosynthesis needs light”
- [ ] A home address

5 **Deleting the original file also deletes what a model learned from it.**

- ◯ True
- ◯ False

✎ **What pupils' data does your school hold that should never train a model? If a service asked for it, what questions would you ask first?**